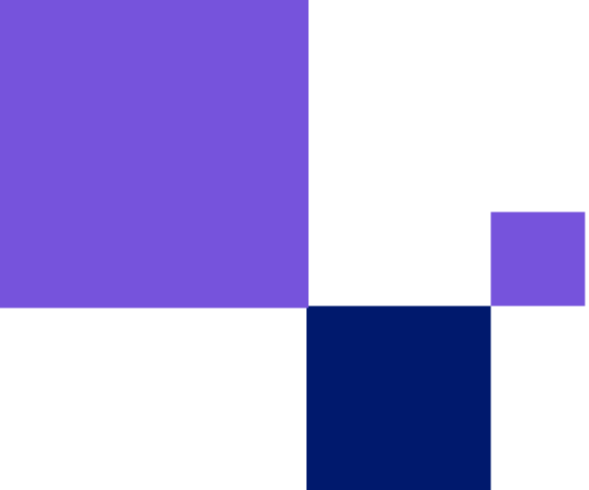
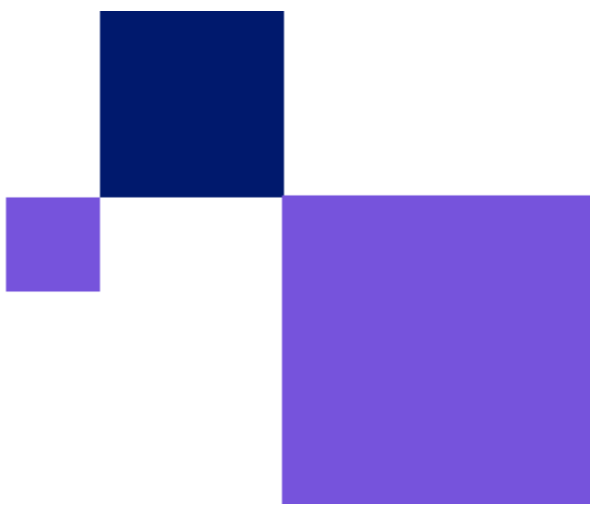




Foundations of Financial Accounting (ACC107)

Lecture 5
Depreciation and Non-current assets
(Chapter 14)



Learning outcomes

By the end of this lecture and with your own self study you should be able to:

- Describe the nature of and valuation of non-current assets
- Discuss the nature of depreciation
- Describe and calculate depreciation using the straight line, reducing balance and sum of years digits methods and describe the circumstances in which each might be the most appropriate
- Show the double entries for depreciation and the sale of a non-current asset
- Calculate the profit or loss on the sale of a non-current asset

Current Assets versus Non-current Assets

Current Assets

- Can be converted into cash within one year
- Asset primarily held for the purposes of trading
- These are assets that are expected to be realized in the entity's normal operating cycle
- Examples: Inventories, trade receivables, short term investments, bank and cash

Non-current Assets

- Items not specifically purchased for resale
- Assets used in the production or distribution of goods normally sold by the business
- Durable goods that last for more than one year
- The entity intends to keep the asset for more than one year

Classification of non-current assets

Non-current assets include **tangible** assets, **intangible** assets and others.

Tangible – non-current assets that have **physical substance**

- land & buildings; plant & machinery; motor vehicles; furniture, fixtures & fittings; office equipment.
- often subject to depreciation

Intangible – identifiable non-monetary assets **without physical substance**.

- goodwill, patents, trade marks, development expenditure.



IAS16 Property, Plant and Equipment is the accounting standard that contains the accounting treatment for **tangible non-current assets**

IAS 16 – Property, Plant and Equipment – Recognition and valuation of non-current assets

Initial Measurement of tangible non-current assets

According to IAS 16 (Property, Plant and Equipment), tangible non-current assets should **initially be measured at cost (historical cost)**

Measurement subsequent to initial recognition

IAS 16 permits two accounting models:

- **Cost model**
- **Revaluation model**

Initial Measurement - Determine the cost - Historical cost (1/2)

The cost of an item of property, plant and equipment **includes**:

- Its **purchase price**.
- Any **costs directly attributable to bringing the asset to its intended location** and **getting the asset to working condition** including:
 - professional fees (e.g. for land and buildings)
 - delivery charges and installation expenses (e.g. for plant and machinery)
 - the costs of testing whether the asset is working properly

Initial Measurement - Determine the cost - Historical cost (2/2)

The cost of non-current assets must **exclude** :

- The costs of any extended **warranty, maintenance agreement** and **replacement/spare parts** where these have been included in the purchase price
- any **road tax and fuel** included in the purchase price of a vehicle.

=> *Revenue expenditure*

Subsequent Measurement – Cost model vs Revaluation Model

Cost Model

Asset is carried at:

Cost

Less (-)

**Accumulated
Depreciation**

Revaluation Model

Asset is carried at:

**Fair value at date of
revaluation**

Less (-)

**Any subsequent
accumulated
depreciation**

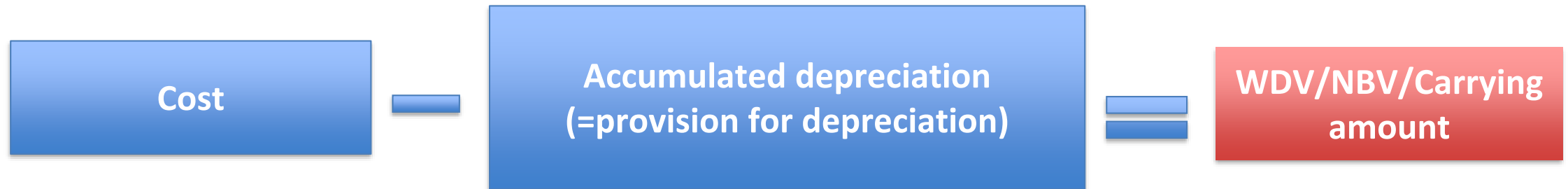
* Note: this topic will be further discussed in “Financial statements of limited company (Lecture 9)”

Subsequent Measurement – Cost Model

Non-current assets' value in SOFP is referred to as “written down value (**WDV**)”, “**net book value (NBV)**”, or “**carrying amount**”

These are exactly the same thing

In the cost model,



Extract from Airbus financial statements for 2023

Property, plant and equipment as of 31 December 2023 and 2022 comprises the following:

	31 December 2023		
	Gross amount	Depreciation / Impairment	Net book value
(In € million)			
Land, leasehold improvements and buildings, including buildings on land owned by others	10,948	(5,783)	5,165

Subsequent Measurement – Revaluation Model Rules

1. Revaluation is allowed when the **fair value** can be measured reliably.
 - *The **fair value** of is the **amount at which the asset can be exchanged** between knowledgeable, willing parties, **in an arm's length transaction***
2. The revalued asset should be **depreciated**.
3. 'Where a tangible non-current asset is revalued **all tangible non-current assets of the same class should be revalued.**' (IASB)
4. 'Revaluations should be made with sufficient regularity to ensure that the carrying amount does not differ materially from that which would be determined using fair value at the reporting date.' (IASB)

** Note: this topic will be further discussed in “Financial statements of limited company (Lecture 9)”*

The Nature of Depreciation (1/2)

- **Depreciation** is a means of allocating the cost of a non-current asset over its useful life in order to match the cost of the asset with the consumption of the assets economic benefits.
- IAS 16 requires all **property, plant and equipment except land** to be depreciated
 - Depreciation is ***NOT*** a means of revaluing non-current assets

The Nature of Depreciation (2/2)

Formal Definitions

- Depreciation is ‘the systematic allocation of the **depreciable amount** of an asset over its **useful life**’ (IAS 16, IASB)
- The **depreciable amount** is ‘*the **cost of an asset**, or other amount substituted for cost, **less its residual value**.*’



- The depreciable amount of a tangible non-current asset should be allocated on a **systematic basis** over its useful life.

How can we estimate the depreciation for each period ?

- ‘The **depreciation method** used should reflect as fairly as possible the pattern in which the asset’s economic benefits are **consumed** by the entity.’

The essential data to compute depreciation (4 determinants)

- **Historical cost/carrying value** of the asset
- **Expected useful life** to the business
- Estimated **residual value** of the asset at the end of its useful economic life.
- **Method of Depreciation**
 1. The straight-line method
 2. The reducing balance method
 3. The sum-of-the-years-digit method

Useful Life

“Useful life is either:

- the **period over which an asset is expected to be available** for use by an entity (number of years); or
- The **number of production** or similar units expected to be obtained from the asset by an entity’ (IASB, IAS 16)”

Residual Value

Residual value is:

‘the **estimated amount** that the entity would currently obtain from the **disposal of the asset**, after deducting the estimated costs of disposal, if the asset were already at the age and in the condition expected **at the end of its useful life.**’ (IASB, IAS 16)

The straight line method

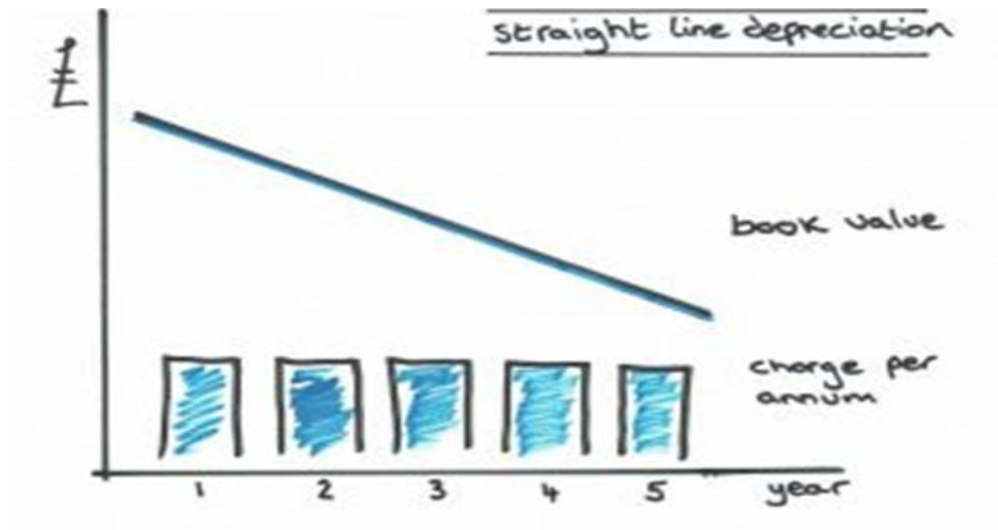
- The **annual depreciation expense**=

$$\frac{1. \text{ Cost - Residual value}}{\text{Useful life (in years)}}$$

or

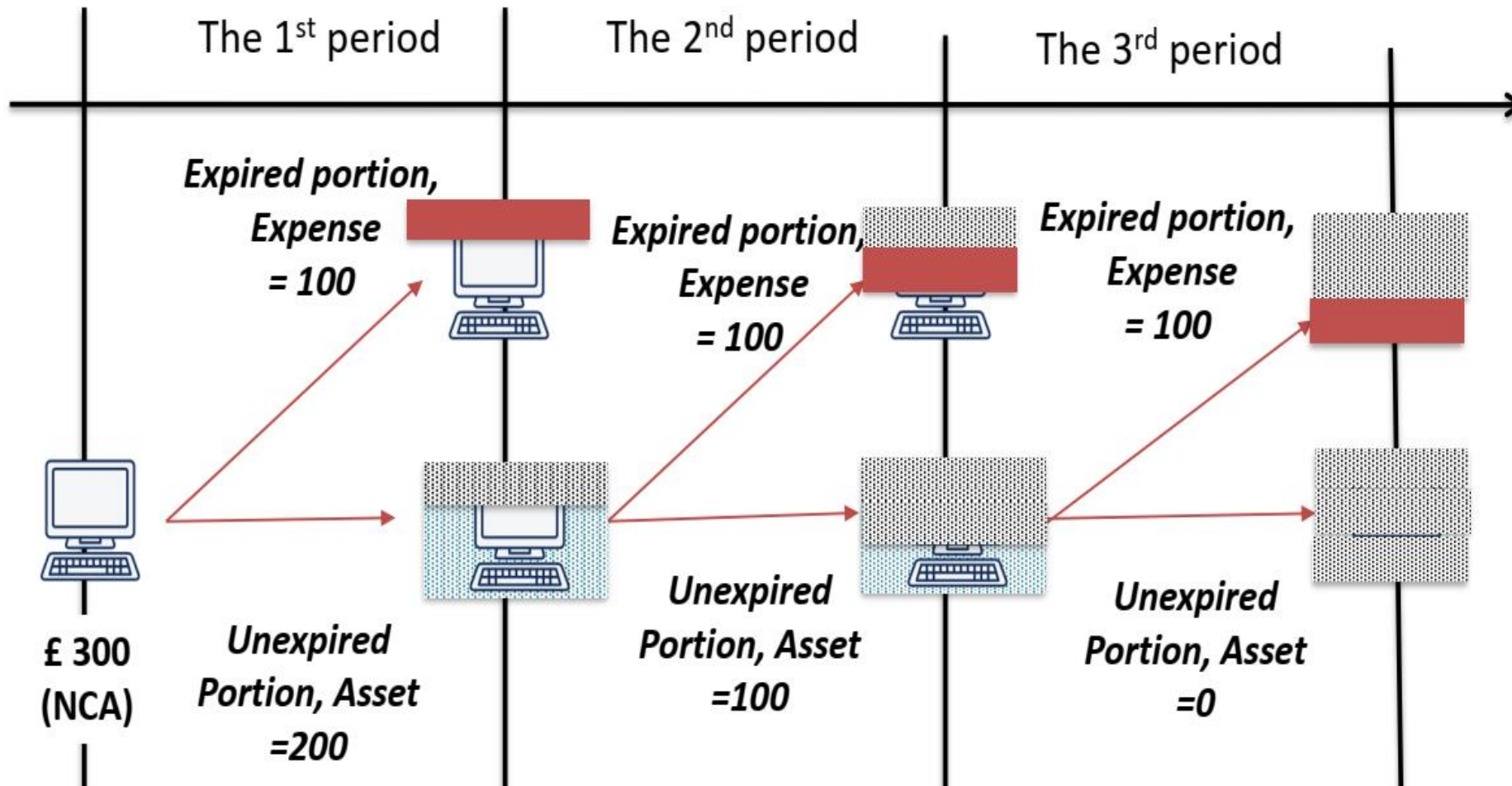
$$2. \text{ Given percentage rate} \times \text{Historical cost}$$

- This gives the same amount of depreciation in each year of the asset's useful life.



The straight line method- Example (1/2)

- Purchase a £300 machine (residual value=0, *useful life* =3 years)
- Assume the **straight-line method** is used and there is no residual value



The straight line method- Example (2/2)

- Purchase a £300 machine (residual value=0, *useful life =3 years*, assuming the revenue= £ 200 each year)

	<i>Cash basis</i>		
	y1	y2	y3
Rev.	£ 200	£ 200	£ 200
Exp.	- £300		
Profit/loss	-£100	+ £200	+200

	<i>Accrual basis (Matching concept)</i>		
	y1	y2	y3
Rev.	£ 200	£ 200	£ 200
Exp.	- £ 100	- £ 100	- £ 100
Profit/loss	+£ 100	+£ 100	+£ 100

- *Which one is more logical for performance evaluation for the given period?*

The Diminishing/Reducing Balance Method

- The percentage rate of depreciation =

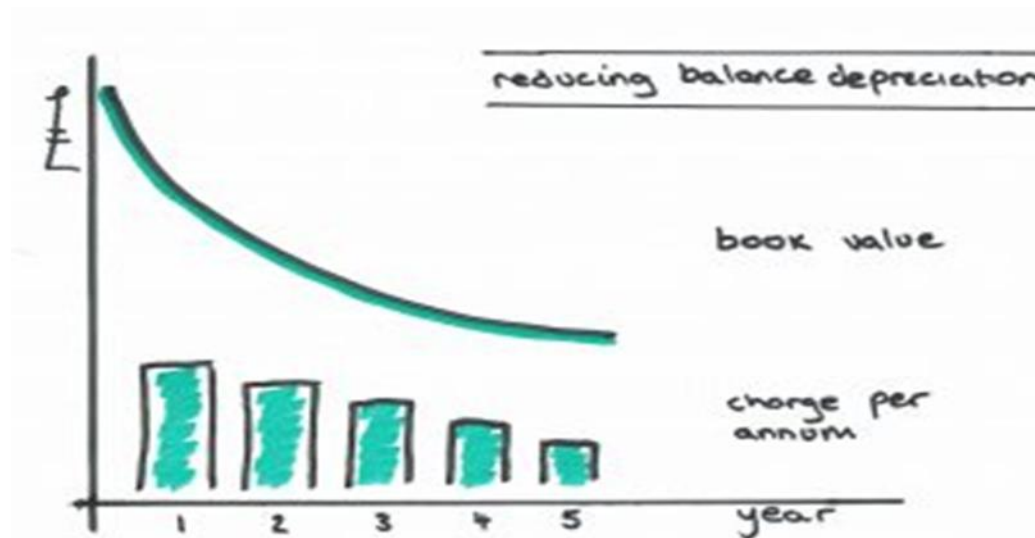
$$\text{Rate of depreciation} = 100 - \left(ul \sqrt{\frac{\text{Residual value}}{\text{Cost}}} \times 100 \right)$$

*ul = useful life

- The annual depreciation expense =

$$\text{Rate (\%)} \times \text{WDV (or NBV) at the start of the year}$$

- This gives a **decreasing annual amount** of depreciation over the assets useful life.



Straight line vs Reducing Balance Method

<i>Straight Line Method</i>	<i>Reducing balance</i>
Suitable for • Most appropriate for assets that are depleted as a result of the passage of time (e.g. buildings). • May be suitable where an asset's utilisation is the same in each year Advantages • It is easy to understand and simple to calculate. Disadvantage • may not give an accurate measure of the loss in value or reduction in useful life (e.g. the large decrease in resale value of vehicles in the first year of their life)	Suitable for • most appropriate for assets that deteriorate as a result of usage where this is greater in earlier years (e.g. plant & machinery, motor vehicles, office equipment). Advantage • It may give a more realistic approximation of the reduction in resale value. Disadvantages • It contains an arbitrary assumption about the rate of decline. • It is relatively complex.

The sum-of-the-years-digits method

- The annual depreciation expense=

$$\frac{\text{Years of remaining useful life at start of year}}{\text{Sum-of-the-years-digits}} \times \text{Depreciable amount}$$

- This gives a **decreasing annual amount** of depreciation over the asset's useful life.
- The arguments for and against are the same as those relating to the reducing balance method but it is **simpler**.

Example 14.1 - Computation of depreciation (1/2)

Plant and machinery that was bought on the 1 January 20X9 at a **cost of £1,000**, has an expected **useful life of 3 years**, and an estimated **residual value of £343**. Calculate the amount of depreciation in each year using:

- the straight-line method;
- The reducing balance method; and
- The sum of the years' digit method

Straight line method:

$$(\text{£}1,000 - \text{£}343) \div 3 = \text{£}219 \text{ per annum}$$

Reducing balance method:

Note: this rate will be given to you in the exam

$$\text{Depreciation Rate} = 30\%$$

$$20X9: 30\% \times \text{£}1,000 = \text{£}300$$

$$20Y0: 30\% \times (\text{£}1,000 - \text{£}300) = \text{£}210$$

$$20Y1: 30\% \times (\text{£}1,000 - [\text{£}300 + \text{£}210]) = \text{£}147$$

$$\text{The NBV at end of 20Y1} = 1000 - 300 - 210 - 147 = \text{£}343 \text{ (residual value)}$$

$$\text{Rate of depreciation} = 100 - \left(ul \sqrt{\frac{\text{Residual value}}{\text{Cost}}} \times 100 \right)$$

Example 14.1 - Computation of depreciation (2/2)

Sum-of-the-years-digits method:

Depreciable amount = $\pounds(1,000 - 343) = \pounds657$

Sum-of-the-years-digits = $3+2+1 = 6$

Annual depreciation:

20X9: $3/6 \times \pounds657 = \pounds329$

20Y0: $2/6 \times \pounds657 = \pounds219$

20Y1: $1/6 \times \pounds657 = \pounds109$

Real World Example of Depreciation Policy - Extract of Tesco's 2024 Annual report

- Most entities provide details of how they value their assets and how they calculate depreciation in their accounting policies.

Real world Example from Tesco plc

Property, plant and equipment

Property, plant and equipment is carried at cost less accumulated depreciation and any recognised impairment in value. Property, plant and equipment is depreciated on a straight-line basis to its residual value over its anticipated useful economic life:

- freehold buildings – 10 to 40 years; and
- fixtures and fittings, office equipment, and motor vehicles – three to 20 years.



The double entries for depreciation

Debit (Dr.)	Depreciation Expense a/c (expense)	£XXX	
Credit (Cr.)	Provision for depreciation /Accumulated depreciation a/c (contra-asset)		£XXX

- The **provision for depreciation** account is referred to as the provision for **aggregate** or **accumulated** depreciation and is deducted from the historical cost to give the WDV

Statement of Financial Position at 31 December 20Y0 (Example 14.1 straight-line depreciation)

<i>Non-current assets</i>			
	Cost	Provision for depreciation	WDV
	£	£	£
Plant and machinery	<u>1,000</u>	<u>438</u>	<u>562</u>

The ledger entries (T-account) under straight line method (textbook) – Figures from example 14.1

Depreciation expense account					
20X9	Details	£	20X9	Details	£
31 Dec	Provision for depreciation A/C	<u>219</u>	31 Dec	P/L A/C (depreciation charge)	<u>219</u>
20Y0			20Y0		
31 Dec	Provision for depreciation A/C	<u>219</u>	31 Dec	P/L A/C (depreciation charge)	<u>219</u>
20Y1			20Y1		
31 Dec	Provision for depreciation A/C	<u>219</u>	31 Dec	P/L A/C (depreciation charge)	<u>219</u>

Provision for depreciation on plant and machinery					
20X9	Details	£	20X9	Details	£
31 Dec	Balance c/d	<u>219</u>	31 Dec	Depreciation expense a/c	<u>219</u>
20Y0			20Y0		
			1 Jan	Balance b/d	219
31 Dec	Balance c/d	438	31 Dec	Depreciation expense a/c	219
		<u>438</u>			<u>438</u>
20Y1			20Y1		
			1 Jan	Balance b/d	438
31 Dec	Balance c/d	657	31 Dec	Depreciation expense a/c	219
		<u>657</u>			<u>657</u>
			20Y2		
			1 Jan	Balance b/d	657

Example 14.1 Extracts from the financial statements under the straight line method – Figures from example 14.1

Extract from SPL: Depreciation Expense		
20X9	20Y0	20Y1
Dep. Expense= 219	Dep. Expense= 219	Dep. Expense: 219
£219	£219	£219

		Partial SOFP		
		Cost	Provision for Depreciation (Accumulated Depr.)	Net Book Value
20X9	Plant & Machinery	£1,000	(219)	£781
20Y0	Plant & Machinery	£1,000	(438)	£562
20Y1	Plant & Machinery	£1,000	(657)	£343*

* Residual value

Disposals of non-current assets

- Profit on sale =

Proceeds of sale $>$ WDV (cost – provision for depreciation)

- Loss on sale =

Proceeds of sale $<$ WDV (cost – provision for depreciation)

Example 14.1 (continued) profit on disposal

		Partial SOFP		
		Cost	Provision for Depreciation (Accumulated Depr.)	Net Book Value
20X9	Plant & Machinery	£1,000	(219)	£781
20Y0	Plant & Machinery	£1,000	(438)	£562
20Y1	Plant & Machinery	£1,000	(657)	£343*

Task

Assume the plant & machinery was sold on the 31 December 20Y1 for £400. Calculate the profit or loss on sale.

$$\text{NBV/WDV} = £1,000 - £657 = £343$$

$$\text{Profit on sale} = £400 - £343 = £57$$

The journal entries for disposals of non-current assets (using example from previous slide)

The **disposal account** is used to make **all** of the entries relating to the sale of the asset and also determines the profit or loss on disposal

Procedure

1. Eliminate the cost of the asset

Debit	Asset disposal a/c	£1000	
Credit	Plant & machinery a/c		£1000

2. Eliminate provision for depreciation (accumulated depreciation)

Debit	Provision for Plant & machinery a/c	£657	
Credit	Asset disposal a/c		£657

3. Account for the sales proceeds:

Debit	Cash or Bank	£400	
Credit	Asset disposal a/c		£400

4. Close the asset disposal account to determine the profit or loss on sale

Debit	Asset disposal a/c	£57	
Credit	Profit or loss a/c (profit on sale)		£57

or

Debit	Profit or loss a/c (loss on sale)	£XX	
Credit	Asset disposal		£XX

The ledger entries for disposals of non-current assets

Plant and Machinery			
20X9		20Y1	
Jan 1	1000	Dec 31 Asset disposal a/c	1,000
Provision for depreciation			
		20Y1	
Dec 31	Asset disposal a/c	Dec 31	Balance b/d
	<u>657</u>		<u>657</u>
Asset disposal account			
20Y1		20Y1	
Dec 31 Plant & machinery a/c	1,000	Dec 31 Bank a/c	400
Dec 31 Profit on sale a/c	<u>57</u>	Dec 31 Prov. For dep. a/c	<u>657</u>
	<u>1,057</u>		<u>1,057</u>
Profit or loss a/c			
Or Dec 31 Assets disposal a/c	<u>XX</u>	Dec 31 Assets disposal a/c	<u>57</u>
(if Dr.>Cr on assets disposal account)			

How is the depreciation expense classified ?

- By **nature** and list **separately** on SPL

– Extract of **BP's 2021 SPL**

	Note	2021
Sales and other operating revenues ^a	5	157,739
Earnings from joint ventures – after interest and tax	15	543
Earnings from associates – after interest and tax	16	3,456
Interest and other income	6	581
Gains on sale of businesses and fixed assets	3	1,876
Total revenues and other income		164,195
Purchases ^a	18	92,923
Production and manufacturing expenses		25,843
Production and similar taxes	4	1,308
Depreciation, depletion and amortization	4	14,805
Net impairment and losses on sale of businesses and fixed assets	3	(1,121)
Exploration expense	7	424
Distribution and administration expenses		11,931
Profit (loss) before interest and taxation		18,082

- By **function** and grouped into different costs (cost of sales, administrative expenses, etc).

– Extract of **Tesla's 2021 Annual report**

Cost of automotive **sales** revenue includes direct and indirect materials, labor costs, manufacturing overhead, including **depreciation costs of tooling and machinery**, shipping and logistic costs, ...

Partial year depreciation*

When a non-current asset is bought or sold part way through an accounting year, the depreciation expense is computed either:

1. on a **strict time basis** – calculate the %

(a) in the year of acquisition, from the date of purchase to the end of that accounting year, and

(b) in the year of disposal, from the start of the accounting year to the date of sale; or

2. if the date of acquisition or disposal is **not given** – normally we assume

(a) a full years charge in the year of purchase, and

(b) none in the year of sale.

**Will not be tested in exams*

Homework

- ▶ Review the content of this lecture
- ▶ Attempt the tutorial questions on time

Thanks for listening

